3.1.2 Amount of substance: Sheet 9

Percentage yield and atom economy · 19 marks · about 25 minutes

Key information

- **% yield = actual amount (or mass) of product ÷ theoretical amount (or mass) × 100**
- Theoretical yield: moles of limiting reactant → use the equation ratio → moles (or mass) of product.
- **% atom economy = M_r of desired product ÷ total M_r of all reactants × 100** (use the balanced equation, including coefficients).
- High atom economy means less waste and fewer by-products to dispose of; it does not depend on the yield.
- Mass of a liquid = volume × density.

Section A: Self-quiz

01 What is the percentage yield when 20 g of aluminium are produced from 50 g of aluminium oxide?

$2Al_2O_3 \rightarrow 4Al + 3O_2$

[1 mark]

A 76%
B 40%
C 33%
D 19%

02 Which reaction has the highest percentage atom economy for the production of hydrogen?

[1 mark]

A $LiH + H_2O \rightarrow LiOH + H_2$
B $CO + H_2O \rightarrow CO_2 + H_2$
C $2\ Al + 3\ H_2O \rightarrow Al_2O_3 + 3\ H_2$
D $CH_4 + H_2O \rightarrow CO + 3\ H_2$

03 What is the percentage atom economy for the formation of sodium nitrate in the reaction between sodium carbonate and nitric acid?

$Na_2CO_3 + 2\ HNO_3 \rightarrow 2\ NaNO_3 + H_2O + CO_2$

[1 mark]

A 36.6%
B 50.3%
C 57.8%
D 73.3%

04 What is the atom economy for the formation of ethylamine in this reaction?

$CH_3CH_2Br + 2\ NH_3 \rightarrow CH_3CH_2NH_2 + NH_4Br$

[1 mark]

A 31.5%
B 35.7%
C 36.1%
D 41.3%

05 A reaction of 1.70 g of methyl benzoate (M_r = 136.0) produces methyl 3-nitrobenzoate (M_r = 181.0) in a 1 : 1 ratio. The percentage yield is 65.0%. What mass, in g, of methyl 3-nitrobenzoate is produced?

[1 mark]

A 0.830
B 1.10
C 1.47
D 2.26

Section B: Exam questions

06 The first stage in the extraction of titanium is the formation of titanium(IV) chloride, $TiCl_4$

$TiO_2(s) + 2\ C(s) + 2\ Cl_2(g) \rightarrow TiCl_4(l) + 2\ CO(g)$

06.1 Calculate the percentage atom economy to form $TiCl_4$ in this first stage of the extraction.

[2 marks]

06.2 Suggest one economic advantage of using a process with a high percentage atom economy.

[1 mark]

07 Copper can be produced by the reaction of carbon with copper(II) oxide according to the equation

$2CuO + C \rightarrow 2Cu + CO_2$

Calculate the percentage atom economy for the production of copper by this process.

Give your answer to the appropriate number of significant figures.

[2 marks]

08 The equation for a process used to manufacture CCl_3F is

$SbF_3Br_2 + CCl_4 \rightarrow CCl_3F + SbF_2Br_2Cl$

Calculate the percentage atom economy for the production of CCl_3F in this reaction.

Give your answer to 3 significant figures.

[2 marks]

09 In an experiment, 10.3 g of a halogenoalkane (M_r = 169.9) is reacted with an excess of ammonia. 2.3 g of an amine (M_r = 59.0) are produced. One mole of halogenoalkane produces one mole of amine.

Calculate the percentage yield in this experiment.

[2 marks]

10 A student prepares some ethyl ethanoate (M_r = 88.0) from ethanol and ethanoic acid. One mole of each reactant produces one mole of ethyl ethanoate.

The student adds 10.0 cm^3 of ethanol (M_r = 46.0) to 5.25 g of ethanoic acid (M_r = 60.0) and obtains 5.47 g of ethyl ethanoate.

For ethanol, density = 0.790 g cm^{-3}

Determine the limiting reagent.

Calculate the percentage yield of ethyl ethanoate.

[5 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	**A**	Source: 7404/1 Jun 2019 Q23	1
02	**D**	Source: 7404/1 Jun 2023 Q12	1
03	**D**	Source: 7404/1 Jun 2024 Q10	1
04	**A**	Source: 7404/2 Jun 2023 Q23	1
05	**C**	Source: 7405/3 Nov 2021 Q29 (1 : 1 ratio stated)	1
06.1	M1 (189.9 ÷ 245.9) × 100 M2 77.2%	Source: 7404/1 Jun 2025 Q01.1–01.2 Allow ECF from incorrect M_r values in M1 Do not allow M2 if ≥ 100%	2
06.2	Idea of maximising the mass of reactants / atoms that ends up in the useful products OR idea of minimising waste / by-product (to be disposed of)	Ignore references to yield and energy Do not allow it costs less unless qualified	1
07	M1 % atom economy = (2 × 63.5) ÷ 171 × 100 M2 = 74.3%	Source: 7404/1 Jun 2017 Q06.4 M2 must be 3 sf	2
08	M1 137.5 ÷ (338.6 + 154.0) × 100 (or 137.5 ÷ 492.6 × 100) M2 27.9(%)	Source: 7404/2 Jun 2022 Q06.1 Can score 1 mark for 137.5 or 492.6 in working if no other marks scored	2
09	M1 amount of amine = 2.3 ÷ 59.0 (= 0.0390 mol) AND amount of halogenoalkane = 10.3 ÷ 169.9 (= 0.0606 mol) M2 % yield = 0.0390 ÷ 0.0606 × 100 = 63.9 to 64(.4)%	Source: 7404/2 Jun 2022 Q05.3 (compound names replaced; 1 : 1 ratio stated) Allow ECF from M1 to M2 Alternative: theoretical mass = 0.0606 × 59.0 = 3.58 g; % yield = 2.3 ÷ 3.58 × 100	2
10	M1 mass of ethanol = 10 × 0.790 (= 7.90 g) M2 amount of ethanol = 7.90 ÷ 46.0 (= 0.172 mol) AND amount of ethanoic acid = 5.25 ÷ 60.0 (= 0.0875 mol) M3 (limiting reagent is) ethanoic acid M4 max mass of ethyl ethanoate = 88.0 × 0.0875 (= 7.70 g) M5 % yield = 5.47 ÷ 7.70 × 100 = 71.0%	Source: 7405/3 Jun 2023 Q01.7 (1 : 1 ratio stated) Allow ECF at each stage; M4 independent of M3 Alternative M4: amount of ethyl ethanoate formed = 5.47 ÷ 88.0 (= 0.0622); M5 = M4 ÷ 0.0875 × 100 M5 allow 70.6 to 71.1 (min 2 sf)	5